

Math for Dotter

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March 12, 2026

Contents

Table of Contents	1
Introduction	3
1 Architecture	4
1.1 The Likelihood Cycle	5
1.2 The Prior Cycle	5
2 The Bayesian Trie	6
2.1 Fields of a Node	6
2.2 Valid Descent	7
2.3 Valid Updates	7
3 Determine Visible Nodes	8
3.1 Setting the Visibility Threshold	8
3.2 Searching for Visible Nodes	8
3.3 Responding to Likelihood Updates versus Prior Updates	8
4 Render Trie	10
5 Stimulus Optimization: Ideal Timer Placement	11
5.1 The Continuous One Bit Channel	11
5.1.1 Two Interpretations	11
5.1.2 The Ideal Timing Distribution	11
5.2 Periodic Signals	12
5.2.1 Random Scan Times	12
5.2.2 Periodic signals	12
5.2.3 Approximating A Uniform Distribution with a Gaussian Mixture	12
5.3 User Parameters	15
5.3.1 The Likelihood Model	15
5.3.2 Directional Statistics	16
6 Render Stimulus	17
7 Detect Response	18
8 Likelihood Result	19
9 Likelihood Update	20
10 Determine Most Likely Break Nodes	21
11 LLM Query	22

12 LLM Result	23
13 Prior Update	24
14 Mastering Dotter	25
14.1 The Cost of Incorrect Signals	25
15 Lazy Updates	27
15.1 Two Kinds of Updates	27
15.2 Likelihood Updates	28
15.3 Prior Updates	28
15.4 Token Paths	29
16 Embellishments	30
16.1 Branches of the Tree	30
Bibliography	30

Introduction

Dotter is a text-entry technology that allows users to type over a 1-bit channel. It allows a simple input mechanism such as blinking or pressing a single button to be used to type. Dotter milks as much information out of that 1-bit channel as possible, allowing an experienced user to type standard english text at speeds of 20 words per minute only by blinking.

Dotter only has one rule: Identify the longest visible prefix of your target text and synchronize your signal to its timer.

Dotter was inspired by the project “Dasher” [2], a text-entry technology developed by David MacKay’s group at the University of Cambridge that took inspiration from arithmetic coding to map the space of all possible text sequences to the unit interval. Assuming a version of Fitt’s law, Dasher is an optimal text-entry technology for 1-dimensional continuous (pointing) input.

This document is a mathematical analysis of Dotter as well as an implementation guide, describing the architecture and key algorithms of Dotter. Chapter 1 describes the architecture of Dotter, outlining its key data structures and algorithms. The succeeding chapters describe individual components of Dotter. The final chapter gives theory and advice on how to learn to type with Dotter. (If the person reading this only wants to learn how to use Dotter, they should skip to the final chapter.)

The name “Dotter” is a nod the most famous 1-bit text-entry system, telegraphy, system which consisted of “dots” and “dashes.” The name “Dasher” was already taken, so I had to go with this.

Chapter 1

Architecture

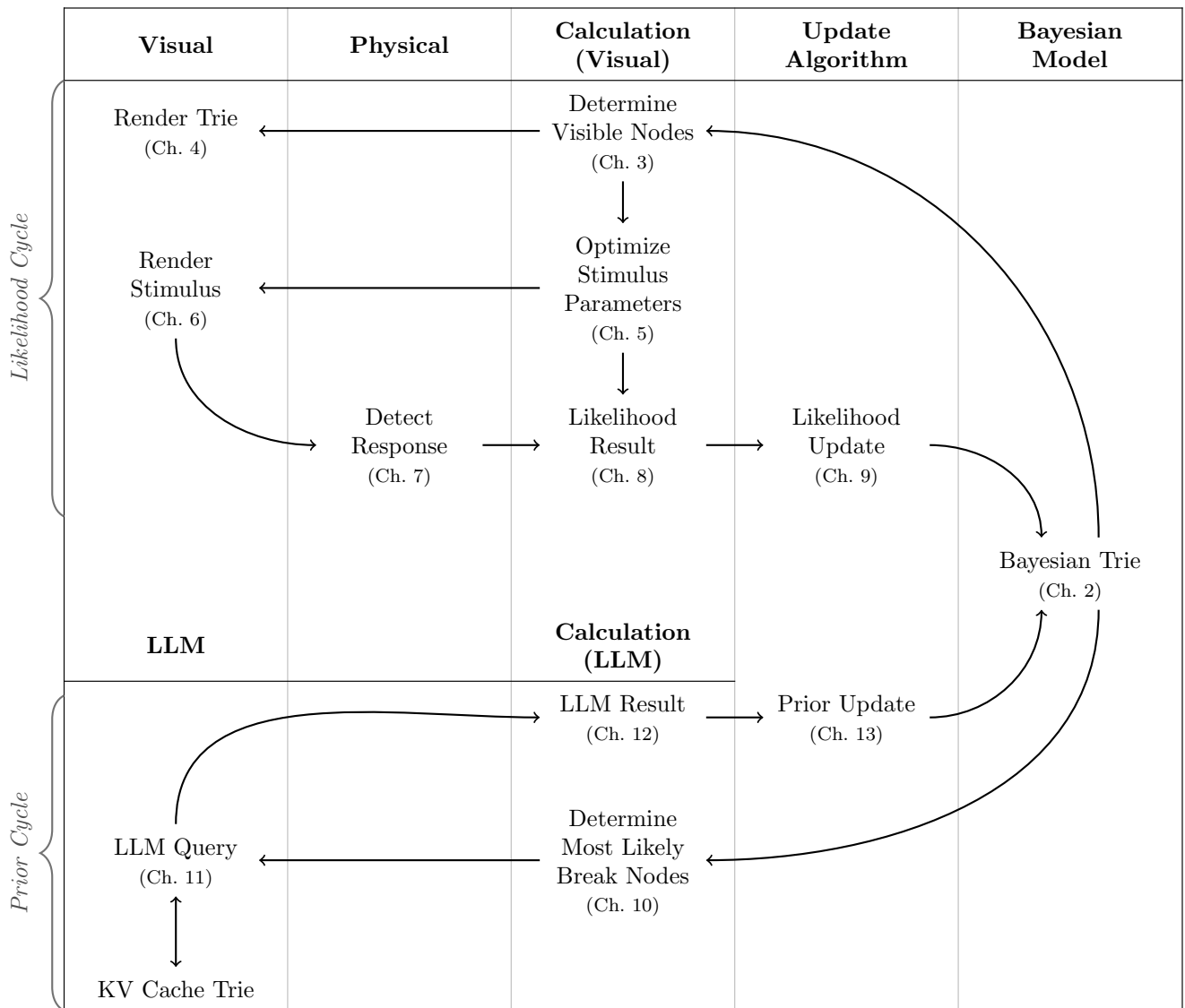


Table 1.1: Architecture of Dotter

Table 1.1 shows the architecture of Dotter. At its heart is the “bayesian trie” (chapter 2) in the rightmost column, which is a Bayesian prefix trie which maintains a calculation of the posterior probability of every prefix. The other elements of the system may be separated into the Likelihood Cycle and the Prior Cycle.

Every element in Table 1.1 is discussed in detail in its own chapter.

1.1 The Likelihood Cycle

The Likelihood Cycle refers to the series of steps by which we elicit a signal from the user and then interpret that signal to update our posterior probabilities. It has seven steps,

1. **Determine Visible Nodes** (chapter 3): we determine which prefixes should be visually displayed by checking which prefixes have a posterior probability exceeding the visibility threshold (e.g. 2%).
2. **Render Trie** (chapter 4): here we draw the visible prefixes to the screen, connecting parent nodes to their children with smooth curves.
3. **Optimize Stimulus Parameters** (chapter 5): in this step we assign timers to every prefix. We want to maximize the information gain from the user’s signal which is accomplished by making the timers as far apart from each other as possible (in a formal sense). This helps to make the user’s selection unambiguous.
4. **Render Stimulus** (chapter 6): we draw the timers to the screen.
5. **Detect Response** (chapter 7): we detect the user’s signal. It is trivial to detect a keypress; detecting the exact timing of a blink is a more sophisticated problem.
6. **Likelihood Result** (chapter 8): we compare the detected user signal to the timers that were assigned in step 3. This gives us a likelihood score for each prefix.
7. **Likelihood Update** (chapter 9): we update the trie to reflect the received likelihoods. Performing this update in a lazy and efficient manner is a central technical challenge of Dotter. (This is an “update” in the traditional bayesian sense.)

1.2 The Prior Cycle

The Prior Cycle refers to the series of steps by which we query a large language model to refine our prior probabilities and incorporate those results into our posterior beliefs. It has four steps,

1. **Determine Most Likely Break Nodes** (chapter 10): we determine which nodes are most likely to a break between tokens in the tokenization of the user’s target text.
2. **LLM Query** (chapter 11): we query the large language model to predict the distribution over the next token for the prefixes identified in the previous step.
3. **LLM Result** (chapter 12): we receive the result from the language model.
4. **Prior Update** (chapter 13): we update the trie to reflect the received prior probabilities. Performing this update in a lazy and efficient manner is a central technical challenge of Dotter. (This is not truly an “update” in the traditional bayesian sense, since normally our priors are fixed.)

Chapter 2

The Bayesian Trie

Dotter must determine the most likely string that the user is typing, based on observation of the user's signals. To accomplish this, Dotter maintains a trie indicating the posterior probability that any given prefix is a prefix of the user's target string.

To compare strings, we use the inequality notation $x \leq y$ to mean that the string x is a prefix of the string y .

Letting X be the user's target string, D be the user's signals, then the posterior probability that a given string, s , is a prefix of X is given by:

$$P(s \leq X|D) = \frac{P(D|s \leq X)P(s \leq X)}{P(D)}$$

In practice, we only keep track of the unnormalized posterior probabilities, denoted in code as `post.Z` (to mean the posterior probability multiplied by a normalizing constant). Since the root node, s_0 , is the empty string, we have $P(s_0 \leq X) = 1$ and thus the unnormalized posterior probability of the root node is $P(D)$, i.e., the normalizing constant. Thus if we know the unnormalized posterior probabilities in our trie, we can convert these to normalized posterior probabilities by dividing by the root node's unnormalized posterior probability.

The posterior probabilities are derived from the prior probabilities and the likelihoods. In Dotter, both are subject to updates. The likelihood is updated as we receive signals from the user and the prior is updated as our language model refines our predictions.

We confront two challenges:

1. How do we update the trie in an efficient and lazy manner, to ensure that the trie is immediately responsive to the user's signals and to results from the language model?
2. How do we efficiently search for visible nodes?

We require two properties of our trie:

1. **Valid Descent:** The root node's unnormalized posterior probability is always correct, and if we descend from the root node to a given descendant node (while possibly performing deferred calculations along the way), then the descendant node's unnormalized posterior probability is also correct.
2. **Valid Updates:** Our algorithms to assimilate likelihood and prior updates preserve the property of valid descent.

2.1 Fields of a Node

A node in the trie has the following fields:

- **letter:** `str` — the character represented by this node.
- **depth:** `int` — the depth of this node in the trie.

- **ever_parent_of_visible_node: bool** — indicates whether this node has ever been a parent of a visible node (at a time when a signal was received). This must be false for the likelihood field to be valid. If this flag is true, the likelihood can only be calculated as a weighted sum of its children’s likelihoods, $P(D|s \leq X) = \sum_l P(D|s + l \leq X)P(s + l \leq X|s \leq X)$, since its children have different likelihoods. Hence, if a node was not ever a parent of a visible node, then its likelihood is known absolutely, but if a node was once a parent of a visible node, then its likelihood is at best known conditional on a prior model. Since our prior model mutates over time, this is not definite knowledge of the likelihood.
- **likelihood: float** — the accumulated likelihood of all signals conditional on this node being a prefix of the user’s target string, $\prod_{d \in D} P(d|s \leq X)$.
- **prior: float** — the prior probability of this node being a prefix of the user’s target string.
- **posterior_Z: float** — the unnormalized posterior probability of this node being a prefix of the user’s target string,
- **unpushed_likelihood: float** — likelihood that has not been accumulated to descendant nodes.
- **break_prior: float** — the prior probability that not only is this node a prefix of the user’s target string, but also that it is the last character of a token in the tokenization of the user’s target string.
- **token_fans: list of floats** — the i -th token fan is the prior probability of this node being a prefix of the target string conditional on its i -th parent being the last character of the token immediately preceding the token to which this character belongs.
- **ancestor_break_priors: list of floats** — the i -th ancestor break prior is the **break_prior** of the i -th ancestor. This may be stale.
- **final_token: str** — the token to which this node belongs, if this node is the last character of a token.
- **final_token_length: int** — the number of characters in the final token.
- **final_token_prior: float** — the conditional prior probability of the final token if the ancestor at depth `depth - final_token_length` is the last character of a token.
- **unpushed_break_priors: list of ints** — a list of depths corresponding to the ancestors of this node whose **break_prior** has changed since the last time we visited this node.
- **unpushed_llm_results: list of ints** — a list of depths corresponding to the ancestors for which we have received next token probabilities from the language model since we last visited this node.

2.2 Valid Descent

The above data structure gives us the machinery to effectively defer the calculation of the posteriors of most nodes.

The posterior for a node `x` is calculated as,

$$x.posterior_Z = x.prior * x.likelihood$$

while the prior is calculated as,

$$x.prior = \sum_{i < x.depth} x.ancestor_break_priors[i] * x.token_fans[i]$$

and the break prior is calculated as,

$$x.break_prior = x.ancestor_break_priors[x.final_token_length] * x.final_token_prior$$

2.3 Valid Updates

Chapter 3

Determine Visible Nodes

In this step we identify which prefixes have sufficient posterior probability to be visually displayed.

The two questions are which prefixes to show and whether to show them immediately or to wait until showing them.

3.1 Setting the Visibility Threshold

The visibility threshold will determine the number of visible nodes, which entails a trade-off.

Advantages of a high threshold and few visible nodes:

- There is less clutter and it is less overwhelming to the user.
- Nodes are larger, avoiding the need for small fonts and timers. The user can recognize timers more easily.
- Faster computation and rendering.

Advantages of a low threshold and many visible nodes:

- More granular targets allows us to spread out the probability more evenly, i.e., we can set the timers to make a better approximation to the uniform distribution. (Refer to chapter 5.)

Thus, we want to set the visibility threshold as high as we can while still being able set the timers to approximate a uniform distribution.

3.2 Searching for Visible Nodes

In order to identify all nodes exceeding the visibility threshold, we descend the trie. We know that posterior probability decreases monotonically with depth, hence we may stop descending upon reaching a node whose (normalized) posterior probability is less than the visibility threshold. It is necessary to descend the trie in the manner described in chapter 2, since this guarantees that the unnormalized posterior probabilities we observe are valid. (Searching for visible nodes with a flat scan over all nodes would be invalid, since many of their posteriors would be stale, since their updates have been deferred.)

3.3 Responding to Likelihood Updates versus Prior Updates

Every time our trie is updated, we need to recalculate the visible nodes. Thus the set of visible nodes will change in response to prior updates as well as in response to likelihood updates. However, these should not be handled in the same way. Since a likelihood update is triggered by the user's gesture, the user expects that the visible nodes will change in response. On the other hand, a prior update is triggered by the background evaluation of a language model. Immediately redrawing in response to these updates would be jarring and disorienting. It could be especially problematic if a child appeared right before the user was about to select

its parent, causing the user to make an error (select the parent instead of the child), through no fault of their own. Similarly, it would be problematic if a node disappeared right as the user was about to select it.

To mitigate this issue, we enforce a couple of rules about redrawing the trie in response to prior updates:

- A new visible node cannot be drawn while its parent is close to culmination.
- A visible node can not be removed until the next likelihood update.

Chapter 4

Render Trie

Chapter 5

Stimulus Optimization: Ideal Timer Placement

5.1 The Continuous One Bit Channel

5.1.1 Two Interpretations

The continuous one bit channel can be conceived in two ways: as a continuous channel with a one bit range, or as a series of times of identical events. The equivalency follows from the fact that the continuous channel may be characterized by the times at which its signal changes. To have a finite (differential) capacity we must constrain the channel to a minimum duration of signal, or equivalently, a minimum time between events. For our purpose, we prefer the latter interpretation of the one bit channel, since the user communicates using a sequence of identical events such as blinks or taps.

5.1.2 The Ideal Timing Distribution

What then, is the capacity of the one bit channel? We prefer the event interpretation for analysis. We consider the ideal distribution of the duration between events. Let T be the minimum duration between events and X be the additional delay beyond the minimum, so that the total duration between events is $T + X$. Let $A = T + E[X]$ be the expected inter-arrival time. The rate is given by

$$C = \max_{\mathcal{P}_X} \frac{h(X)}{A}.$$

We refer to the rate-maximizing distribution of X , $\mathcal{P}_X$, as the ideal timing distribution. We will proceed to show that the ideal timing distribution is an exponential distribution.

We treat the capacity, expected value, and differential entropy as functionals of the probability density function $f(x)$. To find the optimal distribution, we introduce a Lagrange multiplier μ for the normalization constraint $\int f(x)dx = 1$ and define the functional $J(f)$:

$$J(f) = \frac{h(X)}{A} - \mu \left(\int f(x)dx - 1 \right)$$

We maximize $J(f)$ by setting the variational derivative with respect to $f(x)$ to zero:

$$\frac{\delta J}{\delta f(x)} = 0$$

Using the quotient rule for variational derivatives, and noting that $\frac{\delta h(X)}{\delta f(x)} = -1 - \ln f(x)$ and $\frac{\delta A}{\delta f(x)} = \frac{\delta E[X]}{\delta f(x)} = x$, we have:

$$\frac{\delta}{\delta f(x)} \left(\frac{h(X)}{A} \right) = \frac{A(-1 - \ln f(x)) - h(X)x}{A^2}$$

Substituting this into the stationarity condition:

$$\frac{A(-1 - \ln f(x)) - h(X)x}{A^2} - \mu = 0$$

Solving for $f(x)$:

$$f(x) \propto \exp\left(-\frac{h(X)}{A}x\right)$$

proving that X is exponentially distributed with rate parameter equal to the rate of the channel (i.e., $\lambda^* = C$). This fact lets us solve for the rate,

$$C = \frac{1 - \ln C}{T + 1/C}$$

which by algebra has the solution,

$$C = W(T)/T$$

, where W is the Lambert W function.

A Geometric Intuition

The above result (that the ideal timing distribution is exponential) may be appreciated more readily with a geometric argument. We consider a large sample, $\mathbf{X}$, in N -dimensional space, representing the sequence of additional delays X_i . By the law of large numbers, its L_1 norm, $\|\mathbf{X}\|_1 = \sum_i X_i$, must be tightly distributed about the N -simplex, $\|\mathbf{X}\|_1 = NE[X]$. This can be thought of as the permissible region for $\mathbf{X}$. Thus the entropy maximizing distribution for $\mathbf{X}$ should assign a uniform distribution over the N -simplex, which is precisely what the exponential distribution does as it depends only on the L_1 norm, $f(\mathbf{X}) \propto \exp(-\lambda\|\mathbf{X}\|_1)$.

(Incidentally, this conception also provides a geometric proof of the fact that the differential entropy of the exponential distribution is $1 - \ln \lambda$, since the entropy of a uniform distribution over the N -simplex should be the logarithm of its volume. Since its side length is N/λ , its volume is $(N/\lambda)^N/N!$, and its entropy per sample (by Stirling's approximation) is

$$h(X) = \lim_{N \rightarrow \infty} \frac{1}{N} (N(\log(N - \log \lambda) - N(\log N - 1))) = 1 - \ln \lambda$$

as expected.)

5.2 Periodic Signals

5.2.1 Random Scan Times

5.2.2 Periodic signals

5.2.3 Approximating A Uniform Distribution with a Gaussian Mixture

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We have considered the ideal timing distribution and the periodic timing distribution. How can we induce a uniform distribution on the user's signal? There are several targets that the user may aim for, corresponding to all the prefixes displayed. Each target has an assigned signal time, μ_i and a fixed and known (mixing) probability, π_i . Given the noise $\mathcal{E} \sim \mathcal{N}(0, \sigma)$, the distribution of the observed signal is a mixture of gaussians with means μ_i and constant variance σ^2 . Given the mixing probabilities, π_i , how should we position the μ_i to maximize the entropy of the signal?

Intuitively we want to space apart the μ_i to approximate a uniform distribution. Those components with higher π_i should receive more space.

As soon as we receive a signal we recompute the posterior probabilities of the targets and must assign new phases to them. Hence this computation is on the critical path and ought to be computed in a few

milliseconds. This rules out heavier techniques like gradient descent on a numerical approximation of the entropy.

We discuss two heuristic approaches to the problem as well as an approximation for the entropy which may be computed efficiently.

Two Heuristics

UNFINISHED We wish to space the components of the mixture as to approximate a uniform distribution. One heuristic is to determine (by binary search) a value C such that

$$\sum_i 2w_i = P$$

where the widths allotted to each component, w_i , are given by

$$w_i = \text{Re} \left(\sigma \sqrt{2 \left(\ln \left(\frac{\pi_i}{\sqrt{2\pi}\sigma} \right) - C \right)} \right)$$

where P is the length of the interval. The means are then given by $\mu_i = w_i + \sum_{j=1}^{i-1} 2w_j$.

This heuristic is derived from approximating the entropy of the mixture as $h(X) \approx -\ln f_{\min}$ where $f_{\min}$ is the minimum value of the mixture density, and approximating the mixture by its greatest component at a given point,

$$f(x) \approx \max_i f_i(x)$$

The value of C represents the minimum log density of the approximated mixture, i.e., the minimum over the interval of the maximum log density of the components. The width $2w_i$ is the range over which the i th component may provide a log density greater than C . If $\ln f_i(\mu_i) < C$, where $f_i(\mu_i)$ denotes the maximum density of the i th component at $x = 0$, then clearly $w_i = 0$, otherwise it is as given above. Our algorithm thus maximizes the minimum density of the maximum component over the interval. Although somewhat crude, this heuristic is efficient.

A refinement may be made by instead approximating the mixture by the sum of its two greatest components, and continuing to try to maximize the minimum density of the distribution. In this case, adjacent components should be spaced apart such that the log of the sum of their densities attains a minimum value of C . Given two adjacent components, i and $i+1$, we are interested in their separation, l_i . Letting $\mu_i = 0$ and $\mu_{i+1} = l_i$, we may solve for the minimum value of the two-component mixture:

$$f(x) = \pi_i \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right) + \pi_{i+1} \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x-l_i)^2}{2\sigma^2}\right)$$

Note that completing the square for the sum of squared distances gives:

$$(x - \mu_1)^2 + (x - \mu_2)^2 = 2 \left(x - \frac{\mu_1 + \mu_2}{2} \right)^2 + \frac{(\mu_1 - \mu_2)^2}{2}$$

Taking the derivative with respect to x and setting it to zero we have,

$$\frac{d}{dx} f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \left(\frac{-2x\pi_i}{2\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right) - \frac{-2(x-l_i)\pi_{i+1}}{2\sigma^2} \exp\left(-\frac{(x-l_i)^2}{2\sigma^2}\right) \right) = 0$$

Algebra yields,

$$\ln \left(\frac{Rx}{l_i - x} \right) = \frac{x l_i}{\sigma^2} - \frac{l_i^2}{2\sigma^2}$$

, where $R = \pi_i/\pi_{i+1}$. Taken with the condition that $\ln f(x) = C$ the two variables, C and R , implicitly determine l_i . So, using a precomputed table of l_i values for a range of C and R values, we can proceed as before, using binary search to find the value of C such that $\sum_i l_i = P$. As before, we will exclude components with $\ln f_i(\mu_i) < C$ from calculations. As long as each adjacent component has a maximum log density greater than C , it will be possible to space them such that the log of their sum attains a minimum value of C .

Approximations to the entropy based on pairwise distances

In order to optimize the placement of the means to maximize entropy, we may minimize a proxy objective J based on pairwise distances between the components. We consider approximations to the entropy for both Gaussian and von Mises mixture models.

Gaussian Mixture Model with Rényi Entropy The Rényi 2-entropy, also known as collision entropy, is defined as $H_2(X) = -\ln \int p(x)^2 dx$. Maximizing this entropy is equivalent to minimizing the information potential $J = \int p(x)^2 dx$. For a mixture of Gaussians $p(x) = \sum_{i=1}^N \pi_i \mathcal{N}(x|\mu_i, \sigma^2)$, the integral of the squared density is:

$$J = \int \left(\sum_{i=1}^N \pi_i \mathcal{N}(x|\mu_i, \sigma^2) \right)^2 dx = \sum_{i=1}^N \sum_{j=1}^N \pi_i \pi_j \int \mathcal{N}(x|\mu_i, \sigma^2) \mathcal{N}(x|\mu_j, \sigma^2) dx$$

The integral of the product of two Gaussians is itself a Gaussian evaluated at the difference of their means:

$$\int \mathcal{N}(x|\mu_i, \sigma^2) \mathcal{N}(x|\mu_j, \sigma^2) dx = \mathcal{N}(0|\mu_i - \mu_j, 2\sigma^2) = \frac{1}{\sqrt{4\pi\sigma^2}} \exp\left(-\frac{(\mu_i - \mu_j)^2}{4\sigma^2}\right)$$

By dropping the leading scalar and the self-interaction terms ($i = j$) since they are constant with respect to the optimal placement of the means, the objective to minimize simplifies to the pairwise sum:

$$J = \sum_{i < j} \pi_i \pi_j \exp\left(-\frac{(\mu_i - \mu_j)^2}{4\sigma^2}\right)$$

Gaussian Mixture Model with Kolchinsky Objective Kolchinsky and Tracey [1] introduced a family of computationally efficient estimators for mixture entropy that rely on calculating pairwise distances between component distributions, proving they form a strict lower bound on the true mixture entropy. For homoskedastic Gaussian mixtures, maximizing this lower bound utilizing the Bhattacharyya distance yields a similar pairwise objective. The effective variance in the penalty term is simply doubled compared to the Rényi objective:

$$J = \sum_{i < j} \pi_i \pi_j \exp\left(-\frac{(\mu_i - \mu_j)^2}{8\sigma^2}\right)$$

Von Mises Mixture Model with Rényi Entropy When the components are distributed on the unit circle as von Mises distributions with homoskedastic concentration κ , we again aim to minimize the integral of the squared density. The integral of the cross-term of two von Mises components evaluates to a modified Bessel function of the first kind of order zero, I_0 . Stripping out constants and self-terms, the objective becomes:

$$J = \sum_{i < j} \pi_i \pi_j I_0\left(2\kappa \cos\left(\frac{\mu_i - \mu_j}{2}\right)\right)$$

Von Mises Mixture Model with Kolchinsky Objective Applying Kolchinsky's lower bound using the Bhattacharyya coefficient to the von Mises mixture model yields an objective with an identical structure to the Rényi case. The difference lies solely in the concentration parameter, which is halved:

$$J = \sum_{i < j} \pi_i \pi_j I_0\left(\kappa \cos\left(\frac{\mu_i - \mu_j}{2}\right)\right)$$

An approximation for the entropy via Cramér's theorem

UNFINISHED Here we discuss a closed form approximation for the entropy in terms of the mixing weights, π_i , and the chosen means, μ_i . This enables efficient gradient descent on the means μ_i . We take a geometric approach and consider a large sample, $\mathbf{X}$, in N -dimensional space. Let A_i be a discrete random variable

taking on the values of μ_i , with respective probabilities π_i . Let $\mathcal{E}_i \sim \mathcal{N}(0, \sigma^2)$ be random noise on the i th component. Then we have that

$$\mathbf{X} = \mathbf{A} + \mathcal{E}$$

And

$$\begin{aligned} H(\mathbf{X}) &= H(\mathbf{A}, \mathbf{X}) - H(\mathbf{A}|\mathbf{X}) \\ &= H(\mathbf{A}, \mathcal{E}) - H(\mathbf{A}|\mathbf{X}) \quad \text{If } \mathbf{A} \text{ is fixed, } \mathbf{X} \text{ and } \mathcal{E} \text{ differ only by a translation} \\ &= H(\mathcal{E}) + H(\mathbf{A}) - H(\mathbf{A}|\mathbf{X}) \\ &= H(\mathcal{E}) + I(\mathbf{A}; \mathbf{X}) \end{aligned}$$

Since the entropy of the noise is constant, our objective is to maximize the mutual information between $\mathbf{X}$ and the component choice, $\mathbf{A}$. We frame this in terms of trying to infer the true component choice, $\mathbf{A}$, from the observed sample, $\mathbf{X}$. To this end, we consider the prior distribution of the vertices' distances to the sample, $D = \|\mathbf{X} - \mathbf{B}\|_2^2$, where $\mathbf{B}$ represents a candidate component choice and is distributed identically to $\mathbf{A}$. Noting that, $D = \sum_i D_i$ where $D_i = (A_i + \mathcal{E}_i - B_i)^2$, we have that

$$P(\bar{D} = d) \approx \exp(-NI(d))$$

and the likelihood for $\bar{D}$ is

$$P(\mathbf{X}|\bar{D} = d) = \exp\left(-\frac{1}{2\sigma^2}Nd\right)$$

Thus the posterior mode of $\bar{D}$ is attained where $N(I(d) + \frac{d}{2\sigma^2})$ is minimized, which occurs when $I'(d) = -\frac{1}{2\sigma^2}$. At this point, the prior probability of the posterior mode will be $\exp(-NI(d))$, meaning that the observation delivers $NI(d)$ bits of information about the component choice, or $I(d)$ bits per sample.

$I(d)$ had a closed form expression, although its derivation is a somewhat formidable piece of algebra. We begin by defining the random variable $C_i = A_i - B_i$, which is also a discrete random variable taking on the values of $d_{ij} = \mu_i - \mu_j$ with probabilities $\rho_{ij} = \pi_i\pi_j$.

Solving for I in terms of its derivative, and setting it to $-\frac{1}{2\sigma^2}$, we have that

$$\mathbf{I}\left(\theta = -\frac{1}{2\sigma^2}\right) = \frac{1}{2}\ln(\mathbf{Z}) - \frac{1}{4} - \ln(\mathbf{Z}) - \frac{1}{8\sigma^2}\frac{\mathbf{Y}}{\mathbf{Z}}$$

where

$$\mathbf{Z} = \sum_{i,j} \rho_{ij} \exp\left(-\frac{\mathbf{d}_{ij}^2}{4\sigma^2}\right)$$

and

$$\mathbf{Y} = \sum_{i,j} \rho_{ij} \mathbf{d}_{ij}^2 \exp\left(-\frac{\mathbf{d}_{ij}^2}{4\sigma^2}\right)$$

The final term may be recognized as the expected squared distance of C_i under a Boltzmann distribution.

5.3 User Parameters

5.3.1 The Likelihood Model

We model our observations of the user's signal as being subject to two sources of noise: an outlier probability, ρ , that the signal was unintended, and additive noise on intended signals. We model the unintended signals as uniformly distributed, and the noise as a Gaussian. (We do not require that the noise have zero mean, since the user may be consistently late or early, relative to the intended time.) Hence our likelihood model for the received signal is a uniform-gaussian mixture model, characterized by the outlier probability, ρ , and the parameters of the noise, $\mu_\epsilon, \sigma_\epsilon$.

5.3.2 Directional Statistics

Chapter 6

Render Stimulus

Chapter 7

Detect Response

Chapter 8

Likelihood Result

Chapter 9

Likelihood Update

Chapter 10

Determine Most Likely Break Nodes

One challenge of Dotter is that likelihoods are received with reference to a character-based trie, but priors are received with reference to a token-based trie. A considerable portion of the implementation is devoted to closing that gap.

In the chapter 3 we saw how to identify the character sequence prefixes with the highest posterior probabilities so that we could display them to the user. Here our task is to identify the token sequence prefixes with the highest posterior probabilities, so that we can query the language model for predictions. Just as we solved that problem with a character-based trie, so here we will solve it with a token-based trie. A token-based trie has a very high radix equal to the model's vocabulary size, e.g., 50,257 for GPT-2.

In the letter-based trie (see chapter 2), calculating the likelihoods were trivial, but owing to the token-based nature of the language model, calculating the priors of a character sequence was somewhat involved. Here the situation is reversed. Calculating the prior of a token sequence is trivial, since that is directly what the language model returns, while calculating the likelihood of a token sequence prefix is difficult.

Chapter 11

LLM Query

Chapter 12

LLM Result

Chapter 13

Prior Update

Chapter 14

Mastering Dotter

14.1 The Cost of Incorrect Signals

Expertise in Dotter is about accuracy in three ways: (1) The user must synchronize their signals with the target timers. (2) The user must accurately select the correct prefix. (3) The user must not make involuntary signals (e.g. involuntary blinks).

From the program's perspective, there is virtually no difference between the user selecting the wrong prefix and making an involuntary signal. Both are considered as incorrect signals and both are assumed to have random phase. If the user intended the correct timer and was merely imprecise in synchronizing their signal, we do not consider this to be an "incorrect" signal.¹

Here we will quantify the cost of incorrect signals.

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The user's speed can be calculated from their performance parameters. Here we consider the effect of ρ , the probability of unintended signals. Given an observation, D , the gain in logits for the true hypothesis h is $\ln P(D|h) - \ln P(D|\neg h)$. When the posterior probability of h is small, this gain is equal to the gain in the log posterior probability of h . Let r denote the hypothesis that the signal was unintended, so that $P(r) = \rho$.

We state the following facts:

$$\begin{aligned} P(D|\neg h, r) &= P(D|\neg h, \neg r) = P(D|\neg h) \\ P(D|\neg h) &\approx P(D), \quad \text{when } P(h) \approx 0 \\ P(D|h, r) &= P(D), \quad \text{assuming that } D, h \text{ are independent conditioned on } r \end{aligned}$$

The likelihood under the true hypothesis is

$$P(D|h) = \rho P(D|h, r) + (1 - \rho) P(D|h, \neg r)$$

We consider the case where $P \gg \sigma$ so that the components of the mixture are well-separated. This means that the contribution of the false component is negligible and that we can approximate the likelihood $P(D|h)$ piecewise as follows:

$$P(D|h) \approx \begin{cases} P(D|h, r)P(r) & \text{if } D \sim P(D|h, r) \\ P(D|h, \neg r)P(\neg r) & \text{if } D \sim P(D|h, \neg r). \end{cases}$$

That is, when D is likely drawn from the unintended distribution, $P(D|h) \approx P(D|h, r)P(r)$. When D is likely drawn from the intended distribution, $P(D|h) \approx P(D|h, \neg r)P(\neg r)$.

¹We may safely assume that an involuntary signal has random phase. As for an incorrectly selected prefix, if we assume (perhaps unreasonably) that the incorrect prefix was randomly selected from all prefixes, and we further assume (legitimately) that we have uniformly distributed the prefixes' timers, then such a signal has random phase as well.

$$\begin{aligned} E_{D|r,h} [\ln P(D|h)] &\approx E_{D|r,h} \left[\ln \left(P(D|h,r)P(r) \right) \right] \\ &= \ln \rho - h(D|h, r) \end{aligned}$$

and similarly

$$E_{D|\neg r,h} [\ln P(D|h)] \approx \ln(1 - \rho) - h(D|h, \neg r)$$

The expected log likelihood of the data under the true hypothesis is

$$\begin{aligned} E_D [\ln P(D|h)] &= \rho E_{D|r,h} [\ln P(D|h)] + (1 - \rho) E_{D|\neg r,h} [\ln P(D|h)] \\ &\approx \rho(\ln \rho - h(D|h, r)) + (1 - \rho)(\ln(1 - \rho) - h(D|h, \neg r)) \\ &= -H_2(\rho) - \rho h(D|h, r) - (1 - \rho)h(D|h, \neg r) \\ &= -H_2(\rho) - \rho h(D) - (1 - \rho)h(D|h, \neg r), \quad \text{assuming } D|r, h|r, \text{ independent} \end{aligned}$$

Where H_2 is the binary entropy function.

The likelihood of the alternative hypothesis is

$$E_{D|h} [\ln P(D|\neg h)] \approx E_{D|h} [\ln P(D)]$$

To simplify this term we use its expectation over h . Recalling the fact that,

$$E_a [E_{b|a}[f(b)]] = E_b[f(b)]$$

, we find that this term's expectation over h is $-h(D)$.

So the total expected gain in the logits of our hypothesis is

$$E_{D|h} [\ln P(D|h) - P(D|\neg h)] \approx -H_2(\rho) + (1 - \rho)h(D) - (1 - \rho)h(D|h, \neg r)$$

In our particular model, D is uniformly distributed over an interval of length P and the entropy of D conditional on h is just the entropy of the gaussian noise. Denoting the latter term by H_σ , we have,

$$E_{D|h} [\ln P(D|h) - P(D|\neg h)] \approx -H_2(\rho) + (1 - \rho)(\ln P - H_\sigma)$$

This has a nice interpretation: we gain $\ln P - H_\sigma$ bits of information per intended signal, but the presence of unintended signals makes us pay a penalty in two ways: only $(1 - \rho)$ of the time is our signal intended, and additionally we must expend $H_2(\rho)$ bits to learn whether the signal was intended.

Unfortunately the singularity of the binary entropy function at $\rho = 0$ becomes a practical problem for fast text entry. For example, a seemingly low outlier rate of $\rho = 0.03$ imposes the relatively high cost of $H_2(0.03) \approx 0.19$ bits in addition to the 3% loss that would be naively expected. Under reasonable parameters for an intermediate user (e.g. $\sigma = 25ms$, $P = 700ms$), these combined terms amount to a loss of more than 10% of the channel's capacity. Thus to achieve expert entry speeds, it is necessary to make unintended signals very rarely.

Chapter 15

Lazy Updates

The core of Dotter, is calculating the posterior probabilities of prefixes. These posterior probabilities can be calculated exactly since both the prior, the likelihood, and the normalizing constant are all known. However, the challenge is to make these calculations within milliseconds of the user's signal.

Our program maintains a trie of literal prefixes. Our goal is for our updates to touch as few nodes of the trie as possible.

15.1 Two Kinds of Updates

The purpose of an update is to:

1. Identify all nodes with a posterior probability exceeding the visibility threshold.
2. Calculate the posterior probabilities of these nodes. This is equivalent to calculating the unnormalized posterior probability of these nodes (so long as they are using the same normalizing constant), since the posterior probability of the root node must always be 1, and thus we can identify the normalizing constant from the unnormalized probability of the root node.

In order to identify visible nodes, we must know normalized posterior probabilities, which in turn requires knowing the normalizing constant, i.e., the unnormalized probability of the root node. Hence before we can descend the trie to identify visible nodes, we must first find the unnormalized probability of the root node. Thus our update procedure has two phases:

1. Up Propagation: we choose a complete prefix partition for which we know the unnormalized probabilities, and push their sums to the root of the trie `root_post_Z`.
2. Down Propagation: we descend the trie, visiting a child node if it exceeds the visibility threshold, `post_Z - root_post_Z > visibility_threshold`. To visit a node means to create its children if the node is a leaf, or else recalculate its children's statistics. (Note that when later gathering the visible nodes, we must descend the trie. A flat scan could cause a bug since `post_Z` is not guaranteed to be correct a greater depths. I.e., a deep descendant could have a higher `post_Z` than its ancestors if there is unpushed likelihood along the way.)

The trick is to choose the smallest prefix partition for which we can truly calculate the unnormalized posteriors. The unnormalized posterior is a sum over one's children's unnormalized posteriors:

$$P(x|D) = \sum_i P(x_i)P(D|x_i)$$

In general if one term is reweighted, we will have to recalculate, but there are three cases in which we can avoid recalculation and descent:

1. Neither term (neither likelihood nor prior) changes.
2. One term changes by a constant factor, $\sum_i (Ca_i)b_i = C \sum_i a_i b_i$.

3. One set of terms is reweighted to a known sum, and the other set of terms is a constant value across all indices. $\sum_i a'_i b_i = (\sum_i a'_i / \sum_i a_i) \sum_i a_i b_i$.

In order to find the smallest prefix partition for the Up Propagation step, we consider the set of all nodes that satisfy any of these three cases and then take the minimal elements of this set (resulting in an antichain / frontier of the tree).

We update the posterior probabilities in two ways:

1. **Likelihood updates:** when the user makes a gesture, the likelihoods of the prefixes are updated. This is a traditional update.
2. **Prior updates:** when the language model server returns new or better prior probabilities, we must recalculate the posteriors.

	Likelihood Update	Prior Update
Mathematical Term	$P(D X)$	$P(X)$
Recompute Posteriors	Yes	Yes
Resize Boxes	Yes	Yes
How Many New Nodes?	Dozens	Zero to a few
How to Set Phases	Optimized / gradient descent	Random (or ancestor / minimum placement)
Human-Initiated	Yes	No
Show Immediately	Yes	No — wait for ancestor to finish
Branch of Tree Affected	Entire tree	Single branch

Table 15.1: Comparison of likelihood updates and prior updates.

15.2 Likelihood Updates

For our complete prefix partition we choose the set of all nodes that were leaves in the visible trie, along with all invisible nodes that have a visible sibling. Equivalently, we are choosing the minimal elements of the set of nodes where every node has no visible children (resulting in an antichain).

Since all of each of these nodes' children no visible children, each of these nodes' children's likelihoods should be updated in the same way as this node's likelihood (since they all share the same timer). Since all children's likelihood are updated by the same factor, this satisfies case 2 above.

In implementation we add the log likelihood of the user's gesture conditional on this node's timer to the node's `post_Z` and also add this value to the node's `unpushed_likelihood`. If later we determine that this node meets the visibility threshold and thus decide to visit it (i.e., recalculate its children's statistics), we will add the node's `unpushed_likelihood` to the `post_Z` of each of its children and reset the node's `unpushed_likelihood` to 0.

15.3 Prior Updates

Prior updates are more subtle. To find our prefix partition we descend from the token prefix of the update until reaching either (1) a node that *never* was a parent to a visible node, or (2) a space character.

If we reach a node that was never a parent to a visible node, then we are guaranteed that its children all have the same likelihood and thus case 3 above is satisfied. If we reach a space character, then there is only one possible token path up to this point. If that path does not break on the received token prefix, then there is no change to the prior and we are in case 1 above. If that path does contain the received token, then this node and all its literal children will contain the received token in their paths and thus their priors will all change by the same factor which is case 2 above.

In implementation, we do not track the prior as a single value. Instead each literal node tracks the prior assigned to it by each of its (at most 15) possible token paths. The prior update id, the prior value, and the model strength is stored for each of these paths. Unlike a single scalar `unpushed_likelihood`, we store a set

of `unpushed_prior_update_ids`. When we visit a node, we push this set to its children, and appropriately update the children’s priors. By comparing the new prior with the old prior, we can correctly modify the `post_Z` of the children.

We want to ensure that our prior over prefixes is always a valid prior. That means we should always use a valid prior (not the zero prior) for every next token distribution, even if it hasn’t yet been received from the language model. A reasonable choice of default is either a uniform prior over tokens, or a 100% prior on the terminal token. (Note that the terminal token should *never* be visited since subsequent tokens are nonsensical.) However, this technique would also allow for the use of multiple levels of language model power (e.g. an ngram model, a provisional language model, and a full language model).

15.4 Token Paths

Since Dotter uses a letter-by-letter trie, but language models return token probabilities, we face the task of converting a token model to a literal model.

A given literal prefix corresponds to several token paths. For example, the prefix `sci` could be part of the words with the following tokenizations (using the GPT-2 tokenizer):

- `scientific` (single token),
- `science` (single token),
- `sci-ences` (two tokens),
- `sc-iss-or` (three tokens),

We can see that a literal prefix can correspond to multiple token prefixes (empty string, `sci`, and `sc`), and that for a given token prefix, the literal prefix can be generated from multiple terminal tokens (`science`, `scientific`).

Formally, to convert a token model to a literal model, we take the sum across all compatible token prefixes B , and all compatible terminal tokens T ,

$$P(x) = \sum_{B < x} P(B) \sum_{T, x \leq B+T} P(T \mid B)$$

We use two tricks to efficiently compute this sum:

1. We recognize that the eligible longest token prefixes correspond to the prefixes of the last word, since tokens never contain spaces in the middle of them. Additionally we can limit the values of B to at most 15 ancestor prefixes, since tokens are at most 15 characters long in our tokenizer.
2. We can preconvert the language model’s output of logits to an array of *cumulative* probabilities. This allows us to efficiently compute the alphabetical sum over compatible T , by taking the difference between the cumulative probabilities of the start and end.

Chapter 16

Embellishments

16.1 Branches of the Tree

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